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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: DATA ANALYTICS AND VISUALIZATION (24DSB3101) A.Y. 2026 - 2027

TITLE OF THE PROJECT	Machine Learning-Based Healthcare Data Analytics for Early Disease Risk Prediction	
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Abstract

Healthcare organizations generate a vast amount of patient data every day, including medical history, laboratory test results, demographic information, and clinical measurements. Analysing this data manually is time-consuming and may delay the early detection of diseases. Early diagnosis is essential for improving treatment outcomes, reducing healthcare costs, and enabling timely medical intervention. Therefore, there is a need for an intelligent system that can analyze healthcare data efficiently and accurately predict disease risk.

This project proposes a Machine Learning-Based Healthcare Data Analytics System for Early Disease Risk Prediction. The system utilizes benchmark healthcare datasets containing patient attributes such as age, gender, glucose level, blood pressure, BMI, insulin level, cholesterol, and other clinical parameters. The collected data undergoes several preprocessing steps, including missing value handling, duplicate removal, encoding of categorical features, normalization/standardization, outlier treatment, and class balancing using SMOTE, to improve data quality and ensure reliable model performance.

After preprocessing, Exploratory Data Analysis (EDA) is performed to understand the dataset through descriptive statistics and visualizations such as histograms, boxplots, heatmaps, disease distribution charts, and correlation analysis. Feature engineering and feature selection techniques are then applied to identify the most influential healthcare parameters that contribute to disease prediction while reducing irrelevant or redundant features.

The processed dataset is used to train and compare multiple machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, XGBoost, and CatBoost. Each model is evaluated using standard performance metrics such as Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix. The bestperforming model is selected based on its predictive performance and generalization capability.

To enhance transparency and trust in the prediction process, the project incorporates Explainable Artificial Intelligence (XAI) techniques, specifically SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations). These methods explain how individual patient features influence disease predictions, enabling healthcare professionals to understand the reasoning behind the model's decisions.

Finally, the trained model is integrated into a user-friendly web application developed using Flask, where users can enter patient health information and receive real-time disease risk predictions. The prediction history is stored in a MySQL database, and an interactive dashboard presents healthcare analytics, feature importance, disease trends, and model performance through intuitive visualizations.

The proposed system aims to assist healthcare professionals by providing an accurate, interpretable, and efficient decision-support tool for early disease risk prediction, enabling timely diagnosis, personalized treatment planning, and improved patient care.

Signature of the Guide

HOD